

SPACE TECH · MULTI-LEVEL

Space Tech Program

Connect space systems to problems on Earth.



Foundation to CubeSat/CanSat-style industry project.

🕒 24+ weeks

📦 4 modules

🎓 Professional learning

Explore the learning experience →

Practical learning. Clear progression. Work you can explain.

YOUR NEXT CAPABILITY

What will you be able to do?

A deep-tech path through orbital foundations, embedded systems, spacecraft architecture, communications, avionics, AI/autonomy and mission delivery.



Understand spacecraft systems engineering



Build embedded and communication subsystems



Use simulation and mission tools



Complete a CubeSat/CanSat-style team project



Who this is for

- Engineering students
- Embedded developers
- Space-tech aspirants



Your starting point

- Engineering or strong STEM foundation

Your progression at a glance

FROM CONCEPT TO EVIDENCE



Learn it. Apply it. Explain it.

Move from guided concepts to practical exercises and a coherent piece of work. Use the course resources to revisit difficult ideas and make your reasoning visible.

THE CURRICULUM

A clear route through the subject.

Every module builds towards practical work. The course outline below is aligned with the academy's current program catalog.

1 Space Systems Foundations

Mission context and orbital foundations.

- Space mission lifecycle
- Orbital mechanics
- Space environment
- Systems engineering

2 Spacecraft Subsystems

Power, thermal, structures and attitude control.

- Power systems
- Thermal and structures
- ADCS
- Subsystem interfaces

3 Avionics and Communications

Embedded flight and ground communication.

- Embedded systems
- Sensors and telemetry
- RF and ground stations
- Flight software concepts

4 Autonomy and Mission Project

Integrate and demonstrate a mission.

- AI and autonomy
- Simulation and testing
- Mission operations
- CubeSat/CanSat-style capstone

FROM CONCEPT TO EVIDENCE



Practice with purpose

Define a mission objective, reason about system trade-offs and demonstrate an application of satellite or geospatial data.

BUILD SOMETHING THAT DEMONSTRATES YOUR LEARNING

Mission concept and satellite-data application

Define a mission objective, reason about system trade-offs and demonstrate an application of satellite or geospatial data.



Mission concept and trade study



System and ground-segment diagram



Data application and validation notes



Tools and working methods

Python and Linux

Embedded C/C++

Simulation tools

CAD, PCB and FPGA concepts

Tools are part of the learning workflow, not partner endorsements. Examples may evolve while preserving the learning outcomes.



Delivery

Cohort-based learning · 24+ weeks

Confirm the current schedule, delivery arrangements and any prerequisites before enrolling.



Completion

A completion certificate is available after the required course work and a passing assessment.

Course completion is not a job, placement or funding guarantee.

Your next steps

- 1 Review the curriculum and your starting point.
- 2 Review current enrollment options and delivery details.
- 3 Start learning, keep your evidence and track your progress.

[Open the course experience](#) →

Customer-preview edition. Interactive links open the local academy prototype. Confirm course availability, schedule and commercial terms with the academy.

INSIDE THE LEARNING EXPERIENCE

Build a first-pass mission power budget

A small satellite concept has an intermittent payload, a continuous onboard computer and a transmitter used during ground contact. The team needs to estimate average electrical demand.

Understand the concept

Distinguish peak power from orbit-average power. Multiply each operating power by its duty cycle to estimate its average contribution, then add the loads. Energy is power multiplied by time; keep units visible. A first-pass average does not size the battery by itself: eclipse duration, conversion losses, operating constraints and margins also matter.

Worked example

Illustrative loads:

Payload: 8 W at 25% duty cycle -> 2 W average

Computer: 2 W continuously -> 2 W average

Radio: 10 W at 10% duty cycle -> 1 W average

Total before losses and margins: 5 W average

For 1 hour, that average corresponds to 5 Wh.

Interpret the result

The peak simultaneous load could exceed the average substantially. Document which loads may operate together and evaluate the eclipse case separately. The figures illustrate a method, not a flight-qualified design.

Knowledge check

Why is average power alone insufficient to choose a power-system component?

Explanation: Components must also meet peak loads and operating conditions; energy storage must support the required duration and loss assumptions.

YOUR PRACTICAL ASSIGNMENT

Build evidence, not just familiarity.

Build a first-pass mission power budget



Your task

- Prepare a load table with peak power, duty cycle and operating mode.
- Add an eclipse energy case and make assumptions explicit.
- Explain a trade-off between payload operation and communication demand.



Submission review criteria

- Consistent W versus Wh units
- Peak and average requirements treated separately
- Assumptions, losses and margins clearly documented

Submit

A working artifact or analysis, a short explanation of your decisions, and evidence that addresses each review criterion.

Reflect

Describe one failed approach, how you diagnosed it, and what you would test next. Identify limitations instead of hiding them.

Your project connection

Define a mission objective, reason about system trade-offs and demonstrate an application of satellite or geospatial data.

Use the sample assignment as a stepping stone toward the course project, not as a substitute for the full curriculum.

[Explore the worked lesson](#) →

Proposed teaching sample for curriculum and UX review. Full lesson production, instructor review and delivery scheduling remain separate work.